Name :

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test II – March 2025

Programme	:	B.Tech. (CSE)	Semester	:	WS2024-25
Course	:	Embedded Systems	Code	:	BCSE305L
			Class Nbr	:	CH2024250501701 CH2024250501716 CH2024250501718 CH2024250501712
Faculty	:	Mr. Satheesh Kumar T, Dr. Sritama Roy, Dr. Manigandan M, Dr. Sridhar C	Slot	:	G2+TG2
Time	:	90 Minutes	Max. Marks	:	50

Answer ALL the questions

Q.No.	Sub. Sec.	Questions	Marks	BT Level
1.		A software engineer is working on an image processing system for a smart surveillance camera. The camera captures frames and stores metadata as text logs, which contain repetitive sequences such as timestamps, object detections, and alerts. To optimize storage and transmission, the software engineer decides to implement Lempel-Ziv compression to reduce the size of the metadata logs before sending them to a cloud server. One of the metadata logs from the camera contains the following repetitive string: **MTMTZTMTMTZTMTZMTMTZTMTMTZTMTZ.** Assist the software engineer to perform Lempel-Ziv encoding and decoding techniques before sending the camera metadata to cloud server.	10	L3
2.		A newly hired embedded engineer is undergoing hands-on training in developing an automated headlamp system. As their first exercise, they are required to make an LED blink for a specific duration. To accomplish this, the engineer has been provided with the Viper Lite development board. Mention the steps involved in developing embedded software to achieve this task as required by the engineer. Additionally, create a flowchart to visually represent these steps and explain the difference between embedded software development and general software development in C.	10	L4
3.		Embedded KIOSK system was developed to automate the processing of Gratuity Fund (GF) and Public Provident Fund (PPF) ensures accurate computation and timely disbursement of retirement benefits. The employee must work 5+ years for GF, while PPF starts from day one. GF = Last Salary × Years of Experience × 57% PPF = 18% of Salary (monthly from joining) GF: Withdrawable anytime. PPF: Withdrawable only on retirement; transferred if resigned. Draw a suitable program model that could be used for concurrent and sequential processes with merging and synchronization characteristics to achieve the below	10	L4

conditions.

1. The computed gratuity and PPF amounts are forwarded to the finance team for approval. If any discrepancies are found, the request may be sent back for recalculation or verification.
2. Upon receiving finance approval, the system processes the transfer of the calculated gratuity and PPF balance to the employee's registered bank account. Confirmation notifications are sent to the employee.
3. Once the payment is successfully transferred, the system updates its records, marks the process as completed, and generates necessary reports for compliance.

4.	For the mentioned set of tasks, classify them based on the execution pattern and appropriate type of RTOS that could be seamlessly used to service these tasks. Also justify your claims. (i) Sampling ECG signal for every 400ms (ii) Handling incoming network packet in a router. (iii) Emergency braking system in autonomous vehicles (iv) Intruder alert in a security systems (v) Push button in an embedded system (vi) Refreshing display screens at 30ms. (vii) A smart home assistant recognizing human voice. (viii) A firewall detecting threats and responding to cyber-attack. (ix) Engine control unit checking the temperature of engine every 10ms. (x) A smart grid system responding to a power failure in milliseconds.	10	L3
5.	Consider the set of 3 processes whose scheduling parameters are given below in Table.	10	L3

Process No.	Priority	Period (ms)	Scheduling parameters		
			Waiting Time(ms)	Arrival Time (ms)	CPU Burst Time (ms)
P1	2	2	1	5	3
P2	3	4	5	0	1
P3	1	2	2	2	1

Determine the Completion time, Turnaround time. Check under which algorithm the above processes are schedulable and if schedulable draw the gantt chart for the same.